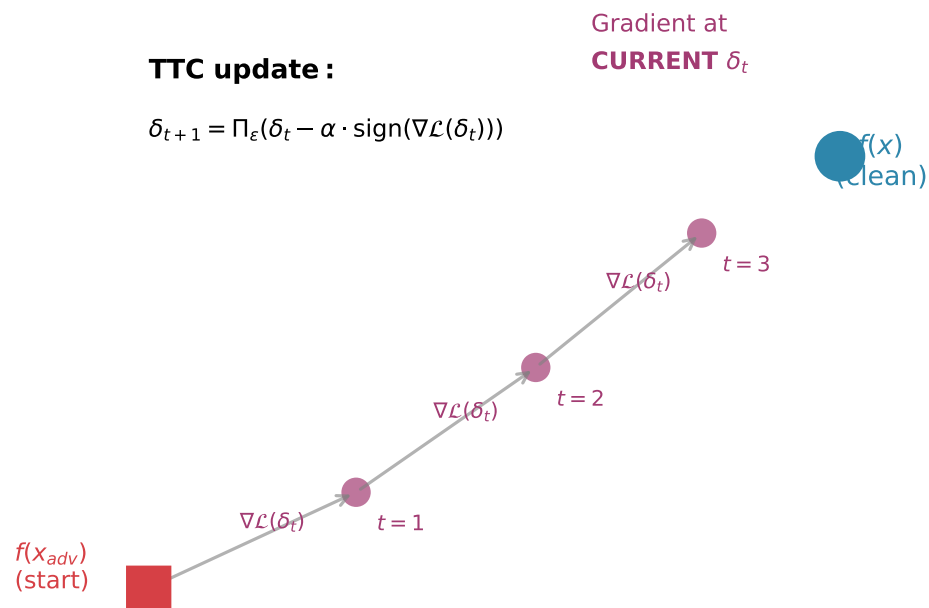
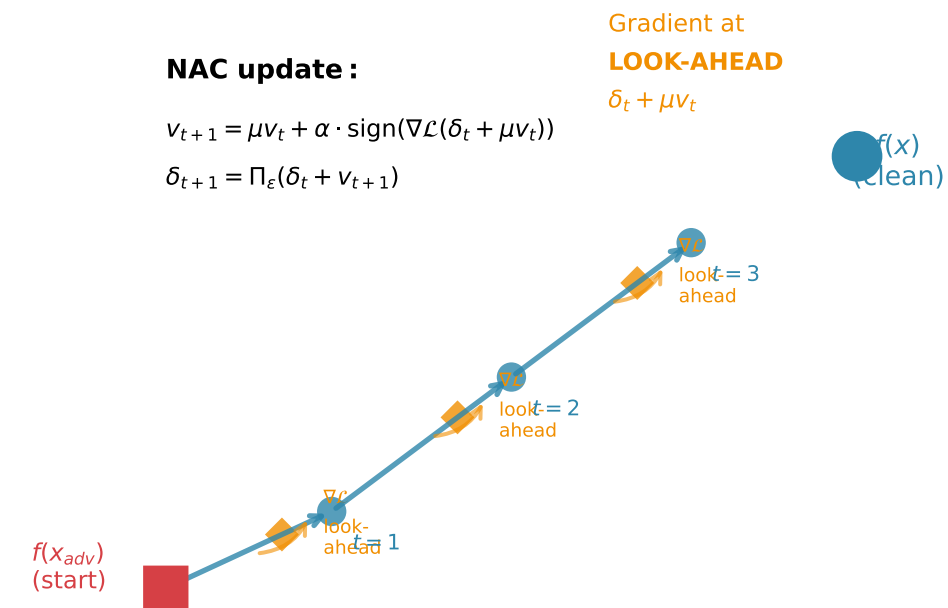


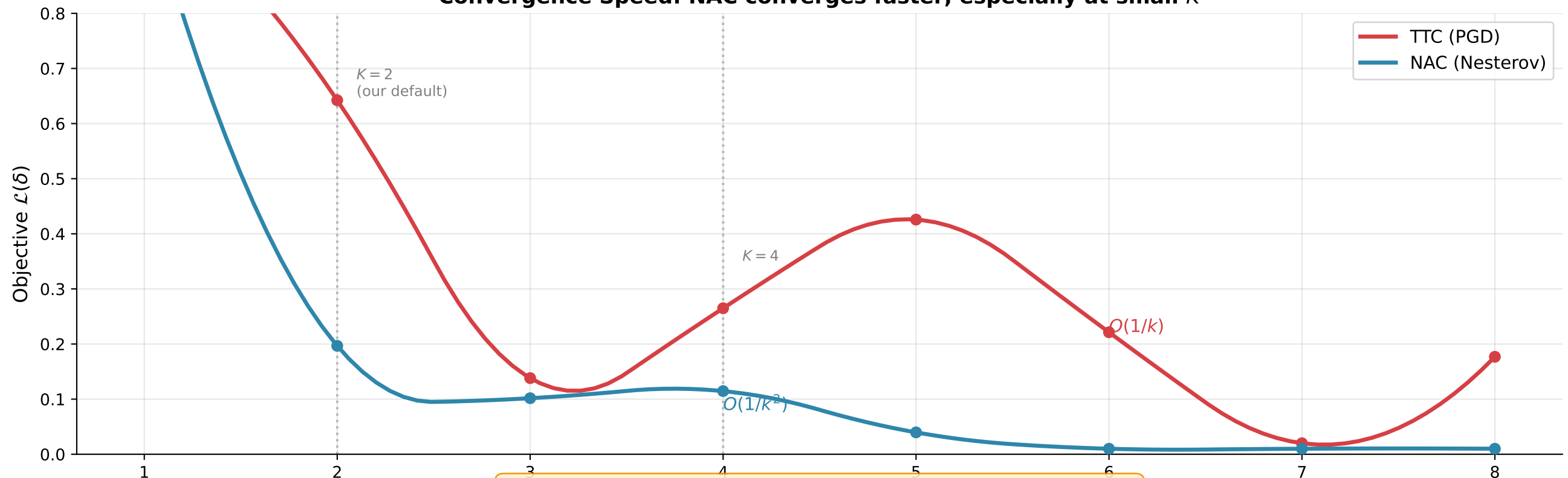
TTC: Gradient at Current Position



NAC: Gradient at Look-Ahead Position (Ours)



Convergence Speed: NAC converges faster, especially at small K



Key Difference: TTC computes gradients at the current perturbation δ_t .
 NAC computes gradients at the look-ahead position $\delta_t + \mu v_t$,
 providing a preview of future gradient directions.